

Project Title:
Statistical Analysis of Smartwatch Data

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1. Data Description

The "Smartwatch Health Data (Uncleaned)" dataset [1], obtained from Kaggle, contains health and activity tracking metrics typically collected from smartwatches devices. It tracks key physiological indicators such as heart rate and blood oxygen levels, alongside behavioral data like daily step counts, sleep duration, activity levels, and subjective stress scores for various users.

The dataset contains 10,000 observations and 7 variables. It exhibits several data quality issues such as missing values, outliers, duplicates, and inconsistencies, indicating that a thorough preprocessing and data cleaning phase is required before any analysis or modeling.

• Data Structure and Variable Types:

Variable Name	Data Type	Description & Noted Quality Issues
User ID	Numeric / Integer	Unique identifier for each user (contains missing values and duplicate records).
Heart Rate (BPM)	Numeric / Float	The user's heart rate measured in Beats Per Minute (contains extreme outliers).
Blood Oxygen Level (%)	Numeric / Float	The percentage of oxygen saturation in the user's blood (contains missing values).
Step Count	Numeric / Float	The total number of steps taken by the user.
Sleep Duration (hours)	Numeric / Float	The total hours of sleep recorded (contains entry errors like "ERROR" strings).
Activity Level	Categorical / Text	The classification of physical activity level (contains formatting inconsistencies and typos).
Stress Level	Numeric / Integer	A numeric score representing the user's subjective stress level.

• Data Exploration:

Several SAS procedures were performed to explore and analyze the data, identify patterns, and detect missing and outlier values.

- Displaying Sample Observations:

```
proc print data=&path (obs=30);  
run;
```

This procedure displays the first 30 observations from the dataset to provide an initial understanding of the data structure and content.

Obs	User ID	Heart Rate (BPM)	Blood Oxygen Level (%)	Step Count	Sleep Duration (hours)	Activity Level	Stress Level
1	4174	58.939775862	98.809650254	5450.3905775	7.167235622316564	Highly Active	1
2	.	.	98.532195159	727.60161005	6.538239375570314	Highly_Active	5
3	1860	247.80305229	97.052953525	2826.5219939	ERROR	Highly Active	5
4	2294	40	96.894213347	13797.338044	7.367789630207228	Active	3
5	2130	61.950165287	98.583796759	15679.067648		Highly_Active	6
6	2095	96.285937686	94.202909778	10205.992256	8.378342673824589	Highly_Active	10
7	4772	47.272256702	95.389760014	3208.7811766	7.871146008904113	Sedentary	2
8	4092	81.733496792	95.981342609	6051.2498567	5.224139066195455	Sedentary	1
9	3979	93.340337681	97.218944847	1775.0295781	5.468413771417153	Sedentary	1
10	3169	51.40996728	97.272786867	7940.021452	7.9544989905339145	Sedentary	8
11	1466	101.91493806	.	2987.5686402	7.541956463173237	Sedentary	9
12	2238	73.852940672	95.399440024	1748.1970688	4.830161671840198	Active	1
13	1330	95.372290367	99.152651594	6155.8373419	5.814348465914493	Highly_Active	10
14	2482	79.062631672	99.014913455	829.92562329	6.269060283034982	Sedentary	
15	3135	68.09020293	99.729739544	8124.1357913	6.444869604525839	Highly_Active	5
16	4444	40	99.047725471	1449.2158953	6.392173806051382	Active	2
17	4171	58.277962699	97.789726451	5643.5113578	7.848859015777246	Sedentary	10
18	3919	.	96.052853484	3488.9195076	6.718661172525927	Sedentary	4
19	4735	98.05924946	96.027099129	5846.9232752	6.621126995655419	Sedentary	9
20	.	53.930206241	96.302569234	1976.1863916	5.3648491527683895	Highly_Active	5
21	2685	74.105002533	99.646064506	2601.3964954	6.751333320225251	Highly Active	9
22	4380	68.310104131	.	1058.3742745	6.456918401449286	Active	3
23	1769	81.267234728	96.068161457	10503.006243	5.780212383241095	Active	
24	3391	76.325520296	98.757110351	508.83809753	4.773999479227033	Sedentary	Very High
25	2515	69.42479492	99.254950764	3306.1104116	8.41542161341735	Highly Active	1
26	4485	84.586624882	99.024686908	4173.3730291	9.169583797303172	Sedentary	9
27	3853	64.627043864	99.78959924	1520.1843869	6.866753232803355	Sedentary	2
28	3433	75.168603508	98.100101284	3400.9009803	8.651900418719752	Sedentary	10
29	2215	86.551598675	97.472634593	3790.0549776	6.844302018581762	Sedentary	1
30	1955	64.104467809	98.147121875	1778.2002773	4.419708490039869	Sedentary	7

The displayed observations revealed several data quality issues, including missing values, inconsistent values, and possible data entry errors.

- Examining Dataset Structure:

```
proc contents data=&path;
run;
```

This procedure was used to display variable names, data types, and other dataset information.

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
6	Activity Level	Char	13	\$13.	\$13.
3	Blood Oxygen Level (%)	Num	8	BEST12.	BEST32.
2	Heart Rate (BPM)	Num	8	BEST12.	BEST32.
5	Sleep Duration (hours)	Char	19	\$19.	\$19.
4	Step Count	Num	8	BEST12.	BEST32.
7	Stress Level	Char	9	\$9.	\$9.
1	User ID	Num	8	BEST12.	BEST32.

The output showed that the variable Sleep Duration (hours) was stored as a character variable instead of numeric, which required data type conversion during preprocessing.

- Summary Statistics:

```
proc means data=&path;  
run;
```

This procedure generates descriptive statistics such as:

- Mean
- Standard deviation (Std)
- Minimum
- Maximum

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
User ID	9799	3007.48	1150.58	1001.00	4999.00
Heart Rate (BPM)	9600	76.0354618	19.4124830	40.0000000	296.5939695
Blood Oxygen Level (%)	9700	97.8415810	1.7328625	90.7912081	100.0000000
Step Count	9900	6985.69	6885.81	0.9101381	62486.69

- Detecting Missing Values:

```
proc means data=&path nmiss;  
run;
```

This procedure was used to calculate the number of missing values in each numerical variable. The results confirmed the presence of missing observations in several variables.

The MEANS Procedure

Variable	N Miss
User ID	201
Heart Rate (BPM)	400
Blood Oxygen Level (%)	300
Step Count	100

```
proc freq data=&path;  
tables 'Activity Level'n 'Stress Level'n;  
run;
```

This procedure displayed the frequency of each category in the Activity Level and Stress Level variable and number of missing or nan values.

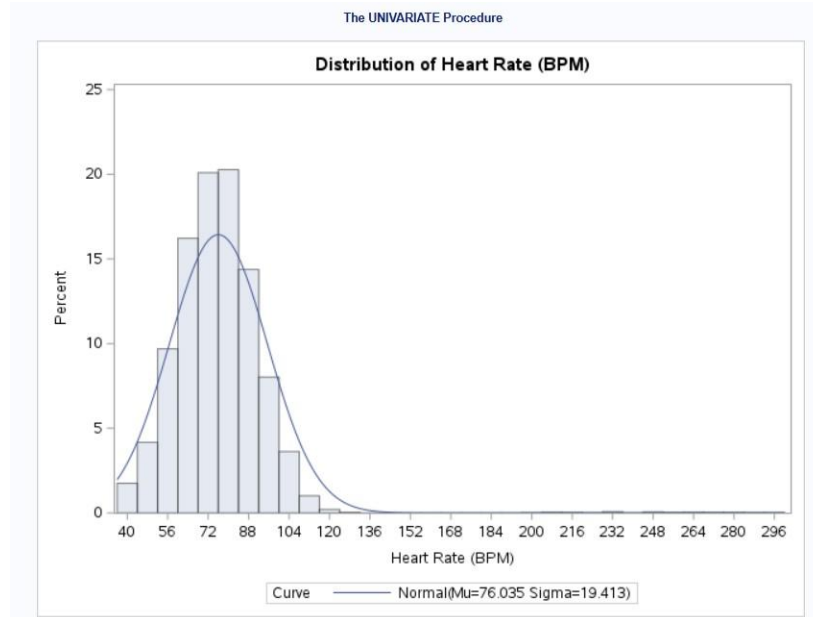
The FREQ Procedure				
Activity Level	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Active	3265	32.65	3265	32.65
Highly_Active	3202	32.02	6467	64.67
Sedentary	3333	33.33	9800	98.00
nan	200	2.00	10000	100.00

Stress Level	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	984	10.04	984	10.04
10	954	9.73	1938	19.78
2	1007	10.28	2945	30.05
3	995	10.15	3940	40.20
4	966	9.86	4906	50.06
5	945	9.64	5851	59.70
6	1001	10.21	6852	69.92
7	1006	10.27	7858	80.18
8	917	9.36	8775	89.54
9	976	9.96	9751	99.50
Very High	49	0.50	9800	100.00
Frequency Missing = 200				

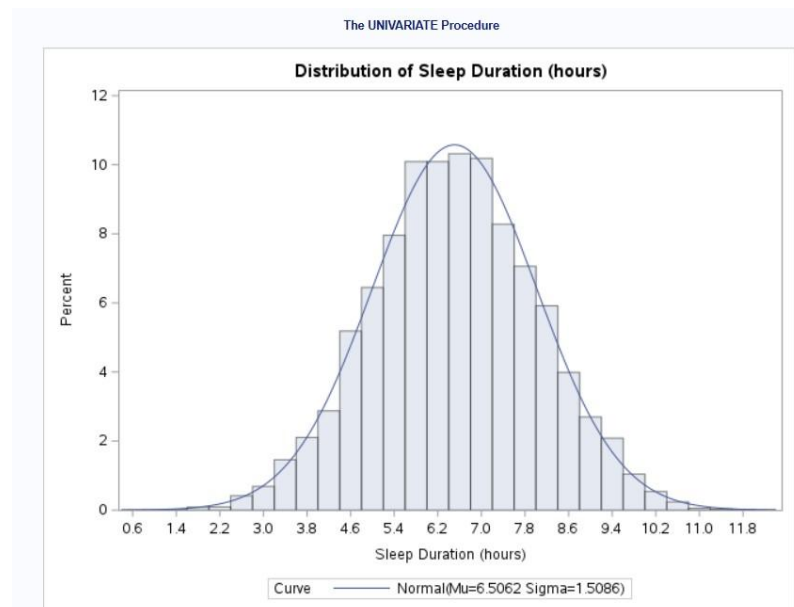
The table shows us that there are spelling inconsistencies and invalid categories such as:

- Active.
 - Sedentary.
 - Very High.
- **Data Distribution:**

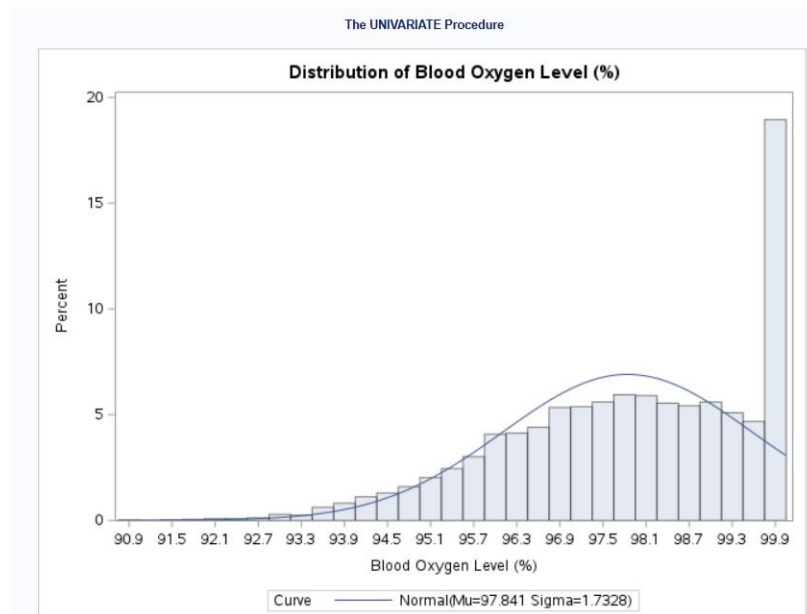
```
proc univariate data=&path normal;
  var 'Heart Rate (BPM)'n 'Sleep Duration (hours)'n
  'Blood Oxygen Level (%)'n;
  histogram 'Heart Rate (BPM)'n 'Sleep Duration
(hours)'n 'Blood Oxygen Level (%)'n / normal;
run;
```



- **Heart Rate (BPM):** The graph indicates that the Heart Rate (BPM) distribution is right skewed and not normally distributed. This skewness is mainly driven by the presence of extreme high values (outliers), which have inflated both the mean and the standard deviation, pulling the distribution toward the right tail.



- **Sleep Duration (hours):** The graph shows that Sleep Duration (hours) is approximately normally distributed. and the distribution appears symmetric with no significant skewness.



- **Blood Oxygen Level (%):** The graph shows a left skewness in the distribution of Blood Oxygen Level (%), along with a clear ceiling effect at the value of 100. This indicates that many observations are clustered at the upper limit, which distorts the normal shape of the distribution and leads to deviation from normality.

2. Data Preprocessing and cleaning:

- **Missing values deletion:**

- During the data exploration phase, the User ID column was found to contain 201 missing values. Since User ID represents a unique identifier for each observation, rows with missing User ID values were removed from the dataset. This step was performed at the beginning of the preprocessing stage to prevent incomplete records from affecting subsequent analyses, including outlier detection and missing value imputation. In addition, records with missing User ID values could not be linked to a specific user and therefore did not provide meaningful analytical value. The following results show the number of missing values before and after deletion.

- **Before deletion:**

```
data &path;
  set &path;
  if missing('User ID'n) then delete;
run;
```

The MEANS Procedure

Analysis Variable : User ID	
N Miss	N
201	9799

- **After deletion:**

```
data &path;
  set &path;
  if missing('User ID'n) then delete;
run;
```

The MEANS Procedure

Analysis Variable : User ID	
N Miss	N
0	9799

- **Correcting typos:**

```
data &path;
  set &path;
  if 'Activity Level'n = 'Active'
  then 'Activity Level'n = 'Active';

  else if 'Activity Level'n = 'Highly Active'
  then 'Activity Level'n = 'Highly_Active';

  else if 'Activity Level'n = 'Seddentary'
  then 'Activity Level'n = 'Sedentary';
run;
```

After applying this procedure, misspellings were corrected in Activity column.

The FREQ Procedure

Activity Level	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Active	3257	32.70	3257	32.70
Highly_Active	3186	31.98	6443	64.68
Sedentary	3320	33.33	9763	98.01
nan	198	1.99	9961	100.00

- **Correcting inconsistencies:**

```
data &path;
set &path;

if 'Stress Level' = 'Very High'
then 'Stress Level' = 10;

if 'Sleep Duration (hours)' = 'ERROR'
then 'Sleep Duration (hours)' = '';
run;
```

This procedure was used to correct data inconsistencies. The value "Very High" in the Stress Level Column was recoded to 10. In addition, invalid entries such as "ERROR" in the Sleep Duration variable were replaced with missing values to allow proper imputation in later steps.

```
proc print data=&path;
where ('Sleep Duration (hours)' is not missing and
'Sleep Duration (hours)' <= 0)
or ('Sleep Duration (hours)' is not missing and
'Sleep Duration (hours)' >= 24);
run;
```

Obs	User ID	Heart Rate (BPM)	Blood Oxygen Level (%)	Step Count	Activity Level	Stress Level	Sleep Duration (hours)
3367	1608	78.832836552	100	7122.8665167	Sedentary	10	-0.19445

This procedure is used to detect inconsistencies in Sleep Duration (hours) column we look for values that are less than 0 or greater than 24 hours and we detect one observation this observation was deleted using below procedure.

```

data &path;
set &path;
if not missing('Sleep Duration (hours)')n and
('Sleep Duration (hours)')n <= 0 or 'Sleep
Duration (hours)')n >= 24)
then delete;
run;

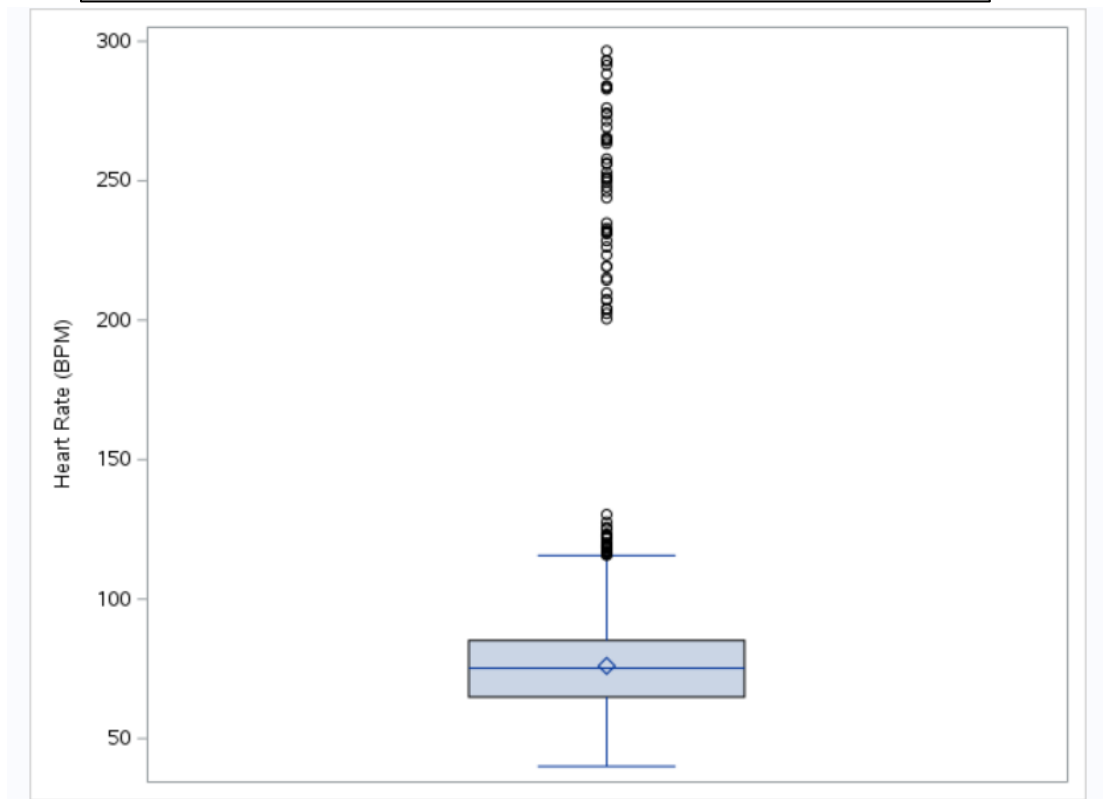
```

- **Outliers' detection:**

```

proc sgplot data=&path;
vbox 'Heart Rate (BPM)';
run;

```



The boxplot indicates the presence of multiple outliers in the Heart Rate (BPM) variable, particularly on the higher end, which contributes to the right skewed distribution of the data.

In this study, potential outliers were identified using the Interquartile Range (IQR) method. According to this method, observations are considered outliers if they fall below the lower bound ($Q1 - 1.5 \times IQR$) or above the upper bound ($Q3 + 1.5 \times IQR$).

A SAS macro was developed to automate the calculation of Q1 and Q3 using PROC UNIVARIATE. The macro computes the interquartile range (IQR) and then derives the lower and upper bounds for each variable. This allows for consistent and efficient detection of outliers across multiple variables.

```
%macro IQR(var, out);
proc univariate data=&path noprint;
  var &var;
  output out=prelib.&out
    pctlpts=25 75
    pctlpre=Q_;
run;
data prelib.&out._final;
  set prelib.&out;
  IQR = Q_75 - Q_25;
  LB = Q_25 - 1.5*IQR;
  UB = Q_75 + 1.5*IQR;
run;
%mend;
%IQR(&hr, HR_iqr);
%IQR(&bo, BO_iqr);
%IQR(&sd, SD_iqr);
```

This procedure is used to compute Q1 and Q3, which are then used to calculate the interquartile range (IQR), lower bound, and upper bound. These values are subsequently used to detect potential outliers.

	Q1	Q3	IQR	Lower bound	Upper bound
Heart Rate	64.908	85.240	20.331	34.412	115.736
Blood oxygen	96.666	99.374	2.707	92.604	103.435
Sleep Duration	5.488	7.529	2.0414	2.426	10.591

Finally, a SQL procedure is applied to extract observations that fall outside the calculated bounds, identifying them as outliers.

- Heart Rate column outliers:

```
proc sql;
select p.*
from &path as p, preplib.HR_IQR_FINAL as h
where p.'Heart Rate (BPM)'n is not missing
and (p.'Heart Rate (BPM)'n < h.LB
or p.'Heart Rate (BPM)'n > h.UB)
order by p.'Heart Rate (BPM)'n desc;
quit;
```

User ID	Heart Rate (BPM)	Blood Oxygen Level (%)	Activity Level	Stress Level	Sleep Duration (hours)	Step Count
4643	296.59396951	95.415587883	nan	10	8.461147	54698
4411	293.2217564	99.52401607	Highly_Active	3	.	11209
1035	292.92329956	94.120618287	Active	4	5.216491	5303
2706	291.40307732	96.239663449	Active	9	7.261535	19209
2283	288.25776091	.	Highly_Active	2	5.291654	21495
2768	284.05525491	96.76196336	Highly_Active	3	7.656777	4723
4902	283.58578538	97.092166156	Sedentary	5	4.823074	11843
1499	283.48314311	96.833301293	Sedentary	9	7.221605	.
4478	282.86312404	97.295126016	Sedentary	3	8.406648	2402
1574	276.12398317	98.386439143	Sedentary	1	9.950313	4255
4797	274.29690416	98.64792509	Highly_Active	1	4.604169	2604
2069	273.86234831	96.509094309	Sedentary	2	7.821971	5002
3856	271.4957856	96.440489265	Sedentary	7	8.514102	841
2655	269.14597467	100	Active	8	6.252117	11331
3156	265.54290951	.	Sedentary	3	4.300171	7281
3381	265.02996728	98.982452508	Highly_Active	6	9.026253	5247
2003	264.64281152	98.242375065	Active	1	8.471285	5299
2129	264.46735634	97.161853495	Highly_Active	7	5.928628	2555
3277	263.37578972	100	Sedentary	1	5.234907	10180
2023	257.82968588	94.316369675	nan	6	4.456317	789
1762	256.37737736	96.026123347	Sedentary	4	7.434144	9969
3218	256.00694521	98.446204905	Highly_Active	10	4.174557	8287
1935	252.99805186	94.263021308	Highly_Active	6	5.452934	25043
4683	251.49052945	97.371765893	Sedentary	9	5.496136	5759
1389	250.84882498	96.558204546	Highly_Active	7	6.249639	2240
4642	250.62067876	97.983059715	Highly_Active	1	7.157805	8526
1844	249.61729139	97.847789802	Sedentary	5	7.103458	1855
1860	247.80305229	97.052953525	Highly_Active	5	.	2826
4629	246.09414448	100	Active	2	6.304732	5614
4976	243.87697796	96.703748174	Sedentary	4	6.272041	4265
4829	234.84093574	97.920202324	Highly_Active	3	5.931396	17207

3200	232.94490693	96.01149176	Highly_Active	9	7.190063	9402
2898	232.12882843	96.082553357	Active	3	5.63989	45
3294	231.69677851	100	Active	9	7.064975	18039
1597	231.25544172	99.396872226	Highly_Active	10	9.7277	23860
3530	231.03628717	99.960102755	Active	4	5.036999	5396
4020	228.49807798	95.280954885	Sedentary	2	9.47121	3794
2008	226.25672509	98.35678919	Highly_Active	.	7.27201	2739
3488	219.28089229	93.422300148	Active	7	6.384323	1167
2443	219.19917821	95.497180737	Highly_Active	9	5.185245	1227
4737	215.19468105	95.095838523	Active	6	6.98847	16072
1698	214.34900887	96.382266411	Sedentary	6	6.801619	3225
2711	209.77392353	97.852780735	Active	5	9.403363	1786
4621	207.43024338	96.688082811	Active	3	8.049854	20577
3261	207.10218769	99.652005742	Active	7	7.218531	2867
3563	204.05760993	95.768379011	Active	3	6.169675	5871
2338	202.41296666	95.948327335	Sedentary	1	9.145513	3935
2369	200.49164217	95.441773037	Highly_Active	9	6.670622	2998
4901	130.31200202	97.044613343	Sedentary	8	.	11532
4679	127.58807105	94.801220072	Highly_Active	6	3.806191	9387
1306	125.73364314	96.788958833	Highly_Active	10	5.298129	791
3509	125.18212037	97.208498446	Active	1	9.931957	2347
1985	123.2547538	98.854288635	Highly_Active	8	6.065564	12990
3399	122.77641723	97.976204127	Sedentary	5	7.378454	8588
3055	122.73567756	100	Highly_Active	2	3.967433	9462
2043	122.10106676	97.793039633	Active	.	5.567505	13174
2560	121.7452205	96.417317402	Sedentary	8	6.478453	2089
2262	120.09870271	98.047939008	Active	9	.	10835
2380	119.90857602	96.858005793	Highly_Active	3	8.988656	2669
3382	119.42489923	99.264651134	Active	1	8.204372	6268
2552	118.94516529	99.177661378	Sedentary	5	9.121201	6489
4410	118.87920563	97.585182036	Sedentary	1	6.750046	572
3352	118.53255952	96.693494273	Active	1	6.901664	5382

2019	118.36817775	98.121702381	Sedentary	6	8.051157	3011
2806	118.29493322	95.629327478	Active	8	5.839286	253
2836	117.71060779	99.308541217	Highly_Active	9	6.185731	10506
2955	117.15926886	97.492693374	Sedentary	9	5.402985	3154
2413	116.88089812	95.349095708	Sedentary	10	8.406885	4705
4854	116.86122514	95.441825211	Sedentary	10	6.798624	4707
4939	116.50548557	99.962976244	Active	3	.	470
4036	116.04667149	99.189513611	Active	1	6.181565	4234

After applying the IQR method, values in the range [116.04, 296.59] were initially identified as statistical outliers. However, these values were further evaluated using medical domain knowledge. According to clinical references [2] [3] [4], heart rate values in healthy individuals typically range between 60 and 100 bpm at rest but can decrease to around 40 bpm in trained athletes and increase above 170–200 bpm during physical activity or specific medical conditions. Therefore, instead of relying solely on the IQR method, a domain-informed threshold was applied, where values below 40 bpm or above 210 bpm were considered physiologically implausible and treated as outliers.

- Sleep Duration outliers:

```
proc sql;
select p.*
from &path as p, preplib.SD_IQR_FINAL as s
where p.'Sleep Duration (hours)'n is not missing
and (p.'Sleep Duration (hours)'n < s.LB
or p.'Sleep Duration (hours)'n > s.UB)
order by p.'Sleep Duration (hours)'n desc;
quit;
```

User ID	Heart Rate (BPM)	Blood Oxygen Level (%)	Activity Level	Stress Level	Sleep Duration (hours)	Step Count
1658	65.173266353	97.361382671	Active	3	12.14023	960
4770	84.63137043	96.004169899	Highly_Active	4	11.90839	9000
3809	74.661369451	100	Active	1	11.65806	10836
2545	60.600080858	96.66236861	Active	3	11.58762	3824
2907	60.751130343	99.876731852	Active	6	11.25837	590
1273	75.10538807	98.688556155	Active	4	11.24858	4948
3003	53.224480595	95.692627864	Active	8	11.1882	.
2963	61.547883344	95.235541917	Sedentary	10	11.11349	5124
1788	91.014160802	97.510911276	Sedentary	6	11.04384	10112
3432	68.374374438	99.941652045	Active	7	11.00769	877
1958	50.827758422	98.787893961	Active	.	10.933	6236
1487	76.834637865	100	Highly_Active	8	10.77184	3952
2750	93.653725808	97.444130405	Sedentary	9	10.75159	6987
1186	63.819276238	100	Active	9	10.70236	2663
3413	87.161390396	99.275651	Sedentary	2	10.69471	13241
2911	85.802511573	98.606868197	Active	6	10.68775	13325
4810	102.1841239	96.217256763	Highly_Active	8	10.65636	7511
1403	59.51176447	98.532708319	Active	8	10.64576	2763
3341	40	99.483077442	Sedentary	4	10.61303	5113
1933	46.363033445	99.736804013	nan	8	10.60905	1510
3265	79.173895479	100	nan	1	10.59606	1492
3469	69.043879778	97.591043389	Sedentary	2	10.59533	4953
1711	69.870240222	96.175939694	Active	9	2.418273	13078
2766	91.591994932	98.450726546	Highly_Active	2	2.39343	8811
1488	80.303351725	96.620155544	Sedentary	4	2.320613	14308
4691	73.656778267	100	Highly_Active	4	2.302539	2044
1976	71.630777431	100	Highly_Active	7	2.285991	4658
4675	92.269195329	98.790524913	Active	8	2.222423	9681
4491	73.087119682	100	Sedentary	3	2.152857	2391
2969	71.510969524	100	Active	.	2.14252	2141
3620	90.118025313	96.12938093	Active	.	2.112415	7204
2874	78.22459114	97.232685454	Highly_Active	5	1.993022	3171
3009	93.576744114	97.150920466	Active	2	1.975132	3377
4078	79.568060978	99.26827044	Highly_Active	2	1.834998	4000
2293	80.840205342	98.691372286	Highly_Active	9	1.809828	5033
3984	74.880180186	97.449861933	Sedentary	5	1.80432	3047
3806	95.094201168	100	Sedentary	9	1.788832	2431
4555	95.76084566	100	Active	8	1.725435	7574
2712	53.079149041	94.937904891	Active	2	1.706087	1955
4401	91.964081888	97.171459765	Active	1	1.402507	652
1343	86.880968249	99.925375676	Highly_Active	7	1.287444	5928
1709	77.09921087	.	Highly_Active	9	1.108614	3465
3936	73.018575346	100	Active	3	0.589987	3546

For Sleep duration (hours), values in the range of [12.140, 0.589] However, only observations below the lower bound were removed, as these values were considered implausible and likely to represent data quality issues. Since the number of such observations was very small (21) observation and had minimal impact on the overall dataset, they were deleted. In contrast, observations above the upper bound were retained because they remained interpretable and logically possible.

- Blood Oxygen Level outliers:

```
proc sql;
select p.*
from &path as p, prelib.BO_IQR_FINAL as b
where p.'Blood Oxygen Level (%)'n is not missing
and (p.'Blood Oxygen Level (%)'n < b.LB
or p.'Blood Oxygen Level (%)'n > b.UB)
order by p.'Blood Oxygen Level (%)'n desc;
quit;
```

User ID	Heart Rate (BPM)	Blood Oxygen Level (%)	Activity Level	Stress Level	Sleep Duration (hours)	Step Count
2722	57.856078083	92.584947021	Active	6	4.483179	14187
2279	59.30440077	92.580824852	Sedentary	10	9.83198	914
4579	72.291004579	92.577289242	Active	5	8.951708	158
1258	97.254386155	92.48812716	Sedentary	8	6.150836	5227
1395	74.03885524	92.483739658	Sedentary	9	5.299209	.
2388	43.259383255	92.4823817	Highly_Active	10	4.913507	994
2150	61.665880504	92.418694924	Active	7	5.590849	20024
3253	52.514029039	92.368105049	Sedentary	1	4.554556	3488
4342	57.523789413	92.28905855	Sedentary	9	6.512628	7163
3914	74.337231042	92.28816657	Active	5	7.455631	2482
1425	43.767313834	92.282996288	Active	1	6.11616	3069
2749	87.110612157	92.19205815	Sedentary	7	4.813027	18406
1110	54.061581055	92.159269892	Sedentary	9	.	15096
1337	40	92.144651616	Active	3	3.281481	5534
1833	78.880652408	92.10938897	Sedentary	4	6.833726	2274
1705	77.218179133	92.009194519	Highly_Active	3	6.419177	3153
4991	89.812695757	92.003999282	nan	7	6.867674	3197
1992	72.889579449	91.973218007	Active	5	7.367893	2367
3073	60.070434254	91.858472777	Highly_Active	1	5.997518	1673
3998	79.414788736	91.818092621	Active	5	8.001758	9767
2094	81.65130076	91.788209693	Sedentary	5	4.650706	7619
1959	75.990920165	91.785861861	Highly_Active	2	5.563478	128
4789	55.749120346	91.717629245	Sedentary	5	5.51493	1539
2265	102.53370657	91.514025941	Active	1	6.769491	1015
3692	89.396976931	91.507534427	Sedentary	8	5.617239	2790
4354	85.644407131	91.461561547	Highly_Active	2	7.078524	5641
3703	84.058395185	91.062167175	Active	4	4.550439	9390
3113	78.345555767	91.034463261	Highly_Active	1	5.688178	15758
2341	61.844490602	90.791208146	Active	3	5.935194	9594

For Blood Oxygen Level (%), values in the range of [90.791, 92.584] as potential outliers. However, these values were retained because they are still medically interpretable and fall within clinically possible ranges. Therefore, although statistically identified as outliers, they were not considered abnormal from a medical perspective and were kept in the dataset.

- Outliers' deletion:

- o Sleep Duration (hours) outliers

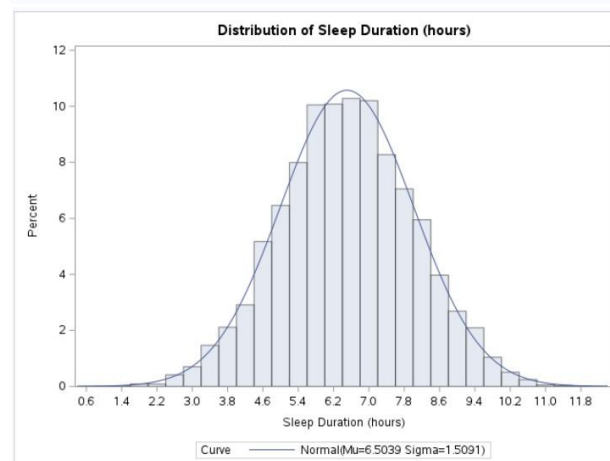
```
data prelib.no_outliers;
set prelib.temp;

if not missing('Sleep Duration (hours)'n)
and 'Sleep Duration (hours)'n < LB
then delete;
run;
```

After identifying outliers and classifying them as either plausible or implausible based on medical evidence, specific data points were removed to ensure data quality. For the Sleep Duration (hours) variable, observations below the lower bound (2.426 hours) were considered unrealistic and removed, resulting in 21 excluded records.

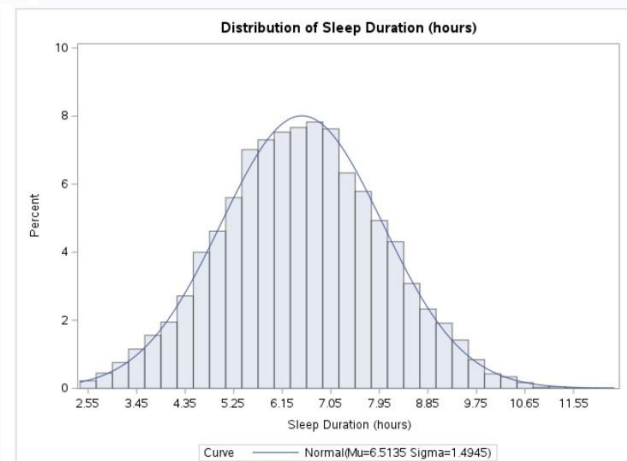
Before imputation:

The UNIVARIATE Procedure Variable: Sleep Duration (hours)			
Moments			
N	9413	Sum Weights	9413
Mean	6.50388628	Sum Observations	61221.0815
Std Deviation	1.50908987	Variance	2.27735223
Skewness	0.01375742	Kurtosis	-0.1007639
Uncorrected SS	419609.391	Corrected SS	21434.4392
Coeff Variation	23.2028944	Std Error Mean	0.01555432



After imputation:

The UNIVARIATE Procedure Variable: Sleep Duration (hours)			
Moments			
N	9352	Sum Weights	9352
Mean	6.51352286	Sum Observations	60914.4658
Std Deviation	1.49449806	Variance	2.23352446
Skewness	0.06058669	Kurtosis	-0.199345
Uncorrected SS	417653.452	Corrected SS	20885.6872
Coeff Variation	22.9445431	Std Error Mean	0.01545408



After removing Sleep Duration outlier values, only minor changes were observed in the statistical measures. The mean, standard deviation, skewness, and kurtosis remained relatively stable, indicating that the overall distribution of the variable was preserved and was already approximately normal before and after outlier removal.

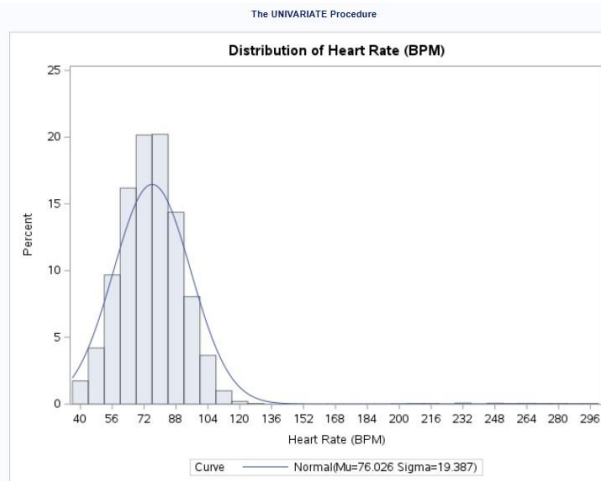
o Heart Rate (BPM) outliers

```
data prelib.no_outliers;
  set prelib.no_outliers;
  if not missing('Heart Rate (BPM)')
    and ('Heart Rate (BPM)' > 210 or 'Heart Rate
(BPM)' < 40)
    then delete;
run;
```

For the Heart Rate (BPM) variable, although some extreme values may be statistically identified as outliers, medical literature indicates that values within a broader physiological range can still be valid. Therefore, only clearly implausible values (heart rates below 40 bpm or above 210 bpm) were removed, resulting in 42 excluded observations.

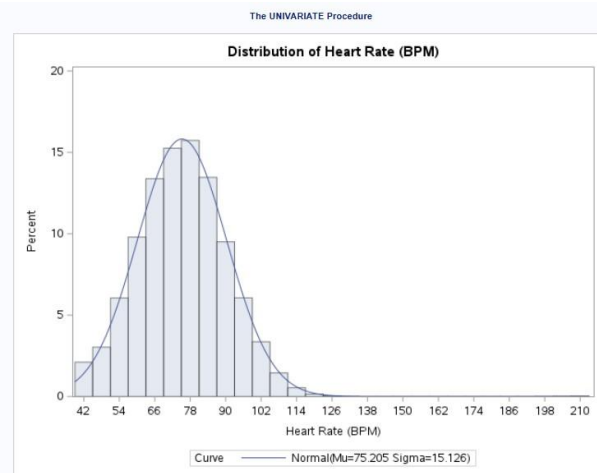
Before imputation:

The UNIVARIATE Procedure Variable: Heart Rate (BPM)			
Moments			
N	9404	Sum Weights	9404
Mean	76.0260063	Sum Observations	714948.563
Std Deviation	19.3872204	Variance	375.864317
Skewness	3.90970065	Kurtosis	36.0116217
Uncorrected SS	57888936.2	Corrected SS	3534252.17
Coeff Variation	25.5007745	Std Error Mean	0.19992138



After imputation:

The UNIVARIATE Procedure Variable: Heart Rate (BPM)			
Moments			
N	9341	Sum Weights	9341
Mean	75.2050018	Sum Observations	702489.922
Std Deviation	15.1263792	Variance	228.807347
Skewness	0.44996887	Kurtosis	3.05897306
Uncorrected SS	54967816.5	Corrected SS	2137060.62
Coeff Variation	20.1135281	Std Error Mean	0.15650863



After removing Heart Rate outlier values, the distribution became less skewed and less affected by extreme values. This is reflected in the significant reduction in skewness, kurtosis, variance, and standard deviation.

- **Missing values imputation:**

The MEANS Procedure

Variable	N Miss
User ID	0
Heart Rate (BPM)	394
Blood Oxygen Level (%)	293
Step Count	97
Stress Level	193
Sleep Duration (hours)	383

To impute missing values, the variables were divided into approximately normally distributed and non-normally distributed data. The Sleep Duration (hours) variable was considered approximately normally distributed, while Heart Rate (BPM), Blood Oxygen Level (%), and Step Count were treated as non-normally distributed variables.

- The **Fully Conditional Specification Regression (FCS REG)** method was used for imputing Sleep Duration (hours), since its distribution was close to normal after outlier removal.
- For the **non-normally distributed variables**, the **Predictive Mean Matching (PMM)** algorithm using the **FCS REGPMM** method was applied. This method was selected because it preserves realistic observed values by matching each missing observation with cases that have similar predicted values and then imputing them from the nearest observed cases. As a result, the imputed values remain plausible and reduce the risk of generating unrealistic or extreme values.

- **SAS Statement Not Taught in Class (PROC MI):**

For missing value imputation, we used **PROC MI** with the **Fully Conditional Specification (FCS)** method, which was not covered in the course. PROC MI is a multiple imputation procedure that models the distribution of missing variables conditional on observed variables, generating statistically sound replacements for missing values.

Within FCS, we applied two sub-methods:

1. **FCS REG** for Sleep Duration (hours) which is approximately normally distributed.
2. **FCS REGPMM (Predictive Mean Matching)** for Heart Rate, Blood Oxygen Level, and Step Count which are non-normally distributed.

- **What is FCS REG:**

This algorithm was used to impute missing values in the Sleep Duration variable. As discussed previously, the Sleep Duration data was approximately normally distributed; therefore, the FCS REG method was selected.

The algorithm works by first building a linear regression model that studies the relationship between the Sleep Duration variable and the other variables in the dataset. The model then predicts the missing values based on these relationships and replaces the missing observations with the predicted values.

Since the variable is approximately normally distributed, linear regression imputation is appropriate and is less likely to generate unrealistic or abnormal values. This helps preserve the original distribution and statistical characteristics of the variable after imputation.

- What is Predictive Mean Matching (PMM):

It is an imputation algorithm used for handling missing data, especially when the data are not normally distributed [5].

How it works:

1. **First:** It builds a linear regression model using observations that do not contain missing values.
2. **Second:** For each missing value, the model predicts an estimated value.
3. **Third:** However, the predicted value is not used directly. Instead, the algorithm searches for the five nearest observed values (called nearest neighbors) whose predicted values are closest to the predicted missing value.
4. **Fourth:** Then, one observed value is randomly selected from these five nearest neighbors and used to replace the missing value.
5. **Finally:** These steps are repeated for all missing values in the dataset.

```
proc mi data=preplib.no_outliers
  seed=12345
  nimpute=1
  out=preplib.imputed_missing_values_data;
  var 'Sleep Duration (hours)'n
      'Heart Rate (BPM)'n
      'Blood Oxygen Level (%)'n
      'Step Count'n;
  fcs reg('Sleep Duration (hours)'n);
  fcs regpmm('Heart Rate (BPM)'n);
  fcs regpmm('Blood Oxygen Level (%)'n);
  fcs regpmm('Step Count'n);
run;
```

The MI Procedure

Model Information	
Data Set	PREPLIB.NO_OUTLIERS
Method	FCS
Number of Imputations	1
Number of Burn-in Iterations	20
Seed for random number generator	12345

FCS Model Specification	
Method	Imputed Variables
Regression	Sleep Duration (hours)
Regression-PMM(K= 5)	Heart Rate (BPM) Blood Oxygen Level (%) Step Count

Missing Data Patterns										
Group	Sleep Duration (hours)	Heart Rate (BPM)	Blood Oxygen Level (%)	Step Count	Freq	Percent	Group Means			
							Sleep Duration (hours)	Heart Rate (BPM)	Blood Oxygen Level (%)	Step Count
1	X	X	X	X	8616	88.51	6.514899	75.206630	97.844444	6990.975390
2	X	X	X	.	86	0.88	6.302539	75.296446	98.001421	.
3	X	X	.	X	270	2.77	6.544332	75.840659	.	6639.149070
4	X	X	.	.	1	0.01	7.213924	107.126196	.	.
5	X	.	X	X	363	3.73	6.515177	.	97.758879	7511.520607
6	X	.	X	.	5	0.05	5.101118	.	98.334556	.
7	X	.	.	X	11	0.11	6.852450	.	.	8218.005247
8	.	X	X	X	353	3.63	.	74.511521	97.970065	6970.927665
9	.	X	X	.	4	0.04	.	73.858796	95.680302	.
10	.	X	.	X	11	0.11	.	77.454324	.	3517.108822
11	.	.	X	X	14	0.14	.	.	97.524480	7553.517315
12	.	.	X	.	1	0.01	.	.	99.462778	.

• Data distribution before & after the imputation:

- **Mean:** The sum of the values divided by their count represents the central or average value of the dataset.
- **Standard deviation:** Refers to the dispersion and spread of the data relative to the mean.
- **Variance:** Is the squared value of the standard deviation.
- **Skewness:** Skewness explores the symmetry or asymmetry of a dataset's distribution. Calculating skewness provides numerical insights into the shape of the data. A skewness value around 0 means symmetric distribution.
- **Kurtosis:** Describe the shape of distribution's level of peakedness or flatness.

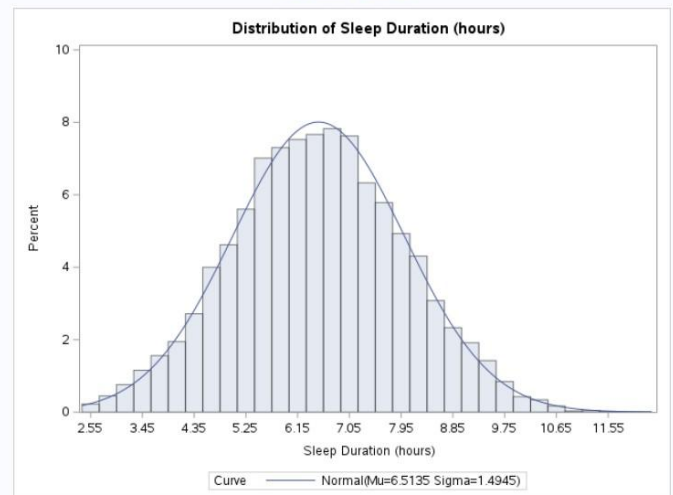
o Sleep Duration (hours)

- **before the imputation:**

The UNIVARIATE Procedure Variable: Sleep Duration (hours)

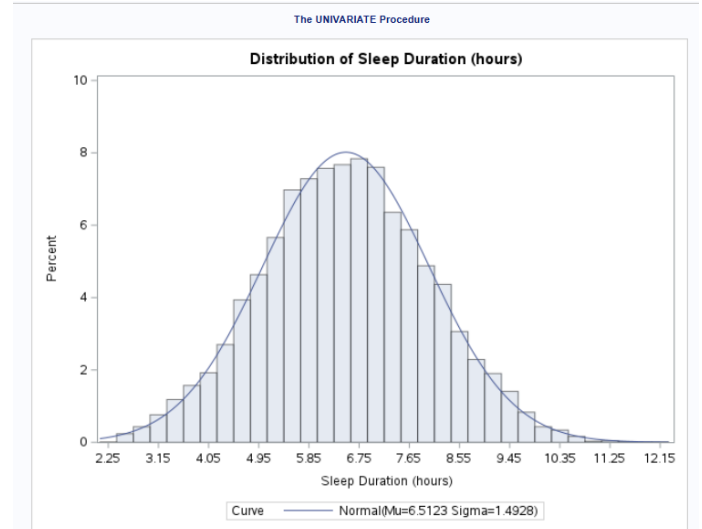
Moments			
N	9352	Sum Weights	9352
Mean	6.51352286	Sum Observations	60914.4658
Std Deviation	1.49449806	Variance	2.23352446
Skewness	0.06058669	Kurtosis	-0.199345
Uncorrected SS	417653.452	Corrected SS	20885.6872
Coeff Variation	22.9445431	Std Error Mean	0.01545408

The UNIVARIATE Procedure



- after the imputation:

The UNIVARIATE Procedure Variable: Sleep Duration (hours)			
Moments			
N	9735	Sum Weights	9735
Mean	6.51233851	Sum Observations	63397.6154
Std Deviation	1.49277559	Variance	2.22837895
Skewness	0.0535045	Kurtosis	-0.1933277
Uncorrected SS	434557.773	Corrected SS	21691.0407
Coeff Variation	22.9222665	Std Error Mean	0.01512957



The table and graph show us the key statistical values mean, variance, standard deviation, skewness, and kurtosis which have changed only slightly, indicating that the data distribution has been preserved.

o Heart Rate (BPM)

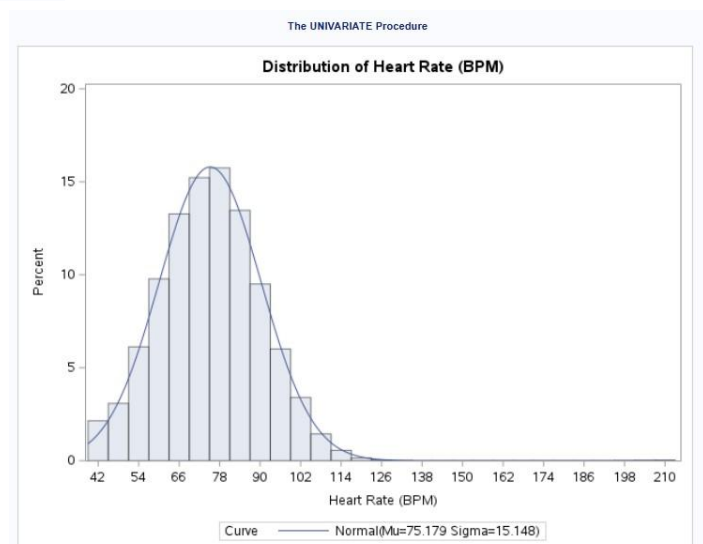
Before imputation:

The UNIVARIATE Procedure Variable: Heart Rate (BPM)			
Moments			
N	9341	Sum Weights	9341
Mean	75.2050018	Sum Observations	702489.922
Std Deviation	15.1263792	Variance	228.807347
Skewness	0.44996887	Kurtosis	3.05897306
Uncorrected SS	54967816.5	Corrected SS	2137060.62
Coeff Variation	20.1135281	Std Error Mean	0.15650863

After imputation:

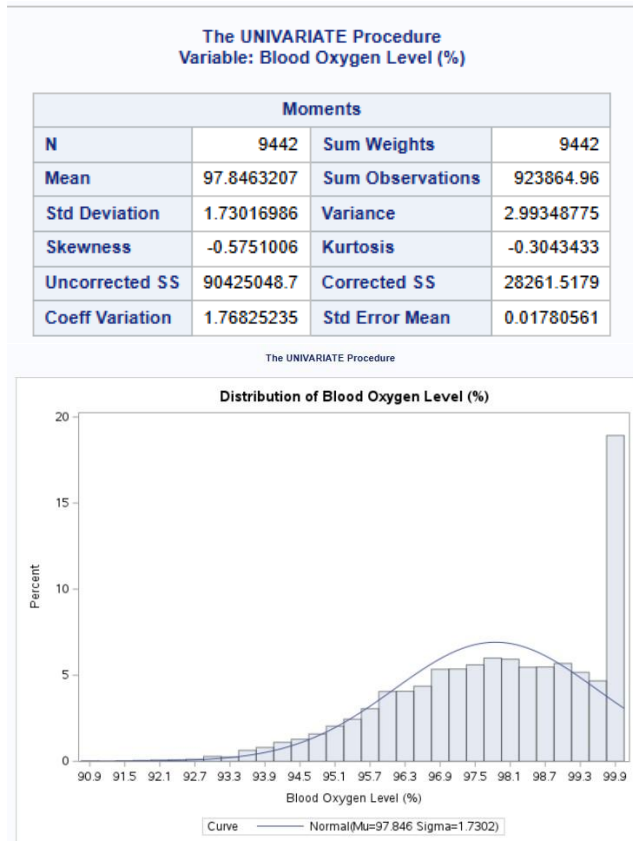
The UNIVARIATE Procedure Variable: Heart Rate (BPM)			
Moments			
N	9735	Sum Weights	9735
Mean	75.1790354	Sum Observations	731867.91
Std Deviation	15.147658	Variance	229.451544
Skewness	0.42963859	Kurtosis	2.90251994
Uncorrected SS	57254604.9	Corrected SS	2233481.33
Coeff Variation	20.1487794	Std Error Mean	0.15352444

Only minimal changes were observed in the mean, standard deviation, variance, skewness, and kurtosis, indicating that the original distribution of the data was successfully preserved after imputation.

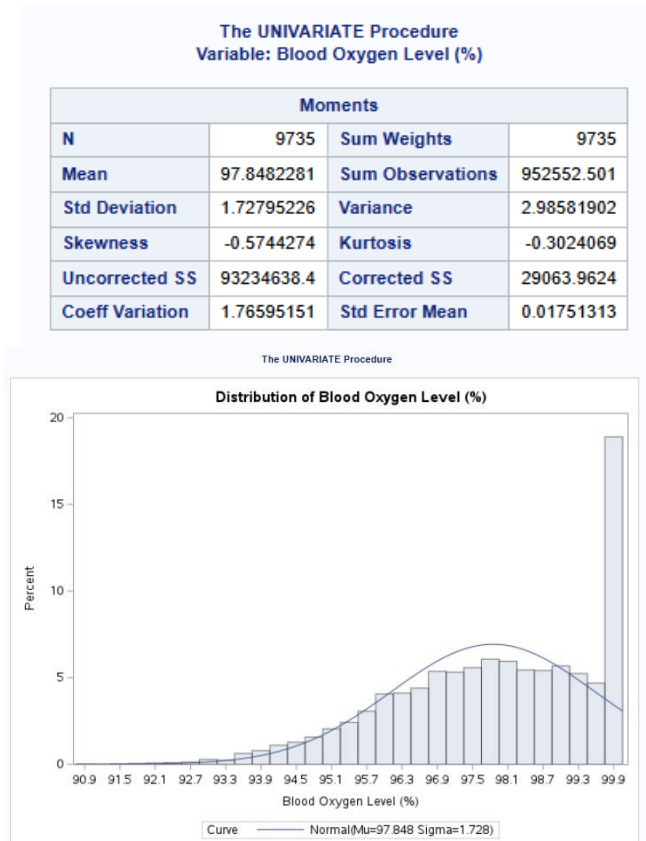


- o Blood Oxygen Level (%)
 - Before the imputation:

Before imputation:



After imputation:



mean, standard deviation, variance, skewness, and kurtosis remained nearly unchanged, indicating that the original distribution and statistical properties of the variable were successfully preserved after imputation.

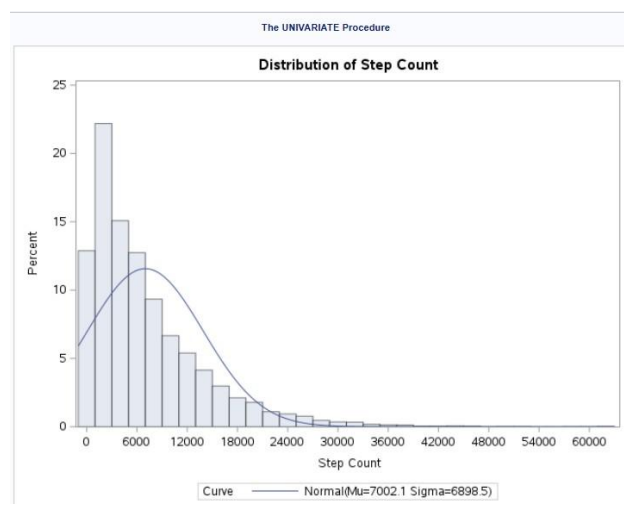
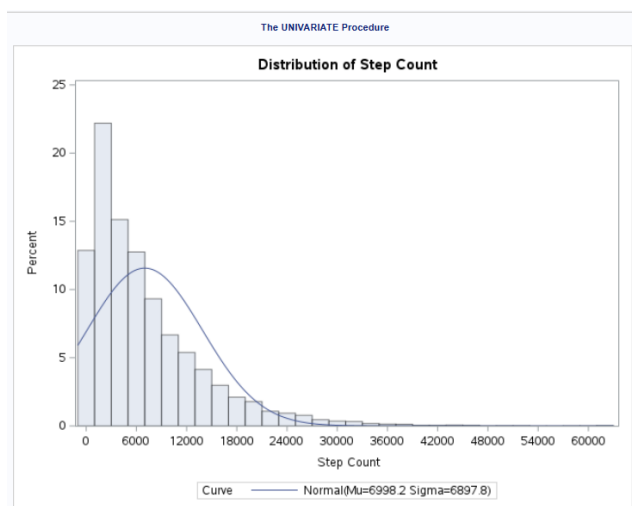
- **Step Count**

Before imputation:

The UNIVARIATE Procedure Variable: Step Count			
Moments			
N	9638	Sum Weights	9638
Mean	6998.24332	Sum Observations	67449069.2
Std Deviation	6897.783	Variance	47579410.4
Skewness	1.95724128	Kurtosis	5.8272741
Uncorrected SS	9.30548E11	Corrected SS	4.58523E11
Coeff Variation	98.5644923	Std Error Mean	70.2612815

After imputation:

The UNIVARIATE Procedure Variable: Step Count			
Moments			
N	9735	Sum Weights	9735
Mean	7002.14883	Sum Observations	68165918.9
Std Deviation	6898.48837	Variance	47589141.8
Skewness	1.94983772	Kurtosis	5.77636182
Uncorrected SS	9.40541E11	Corrected SS	4.63233E11
Coeff Variation	98.5195907	Std Error Mean	69.9175109



- **Stress Level**

The Stress Level variable is an ordinal variable with values ranging from 1 to 10. Therefore, median imputation within each Activity Level category was used instead of regression-based methods. The median was selected because it is more appropriate for ordinal data and is less affected by skewness or outliers. In addition, grouping by Activity Level helps preserve the relationship between activity level and stress level, resulting in more realistic imputed values.

```
Proc means data=preplib.imputed_missing_values_data nway
  median noprint;
  class &al;
  var &sl;
  output out=preplib.medians
  median = Median_Stress;
```

Create new dataset storing median of each stress level by activity level class

```
proc sql;
    create table prelib.imputed_sl_filled
    as select a.*,
    b.Median_Stress
    from prelib.imputed_missing_values_data as a
    left join prelib.medians as b
    on a.'Activity Level'n = b.'Activity Level'n;
```

Create new table that stroe both our dataset and median stress level data

```
data prelib.imputed_sl_filled;
    set prelib.imputed_sl_filled;
    if missing('Stress Level'n)
    then 'Stress Level'n = Median_Stress;
run;
```

Filling missing values

The MEANS Procedure	
Variable	N Miss
User ID	0
Heart Rate (BPM)	0
Blood Oxygen Level (%)	0
Step Count	0
Stress Level	0
Sleep Duration (hours)	0

- Activity Level
First values with (nan) converted into missing values "

```
data prelib.imputed_missing_values_datanona;
    set prelib.imputed_sl_filled;
    if &al = 'nan'
    then call missing(&al);
run;
```

The FREQ Procedure

Activity Level	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Active	3186	33.40	3186	33.40
Highly_Active	3122	32.73	6308	66.12
Sedentary	3232	33.88	9540	100.00
Frequency Missing = 195				

For activity level we tried to predict the missing values using logistic regression model
First, we split the data into train and test data with 70% 30% size

```
proc surveyselect data=prelib.imputed_missing_values_datanona
  out=prelib.spilited_data
  seed=12345
  samprate=0.7
  outall;
run;
data prelib.train prelib.test;
  set prelib.spilited_data;
  if Selected = 1
  then output prelib.train;
  else
  output prelib.test;
run;
```

The FREQ Procedure

Selection Indicator				
Selected	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	2920	29.99	2920	29.99
1	6815	70.01	9735	100.00

```

proc logistic data=prelib.train
  outmodel=prelib.Activity_Level_model;
  class 'Activity Level'n (ref='Sedentary') / param=ref;

  model 'Activity Level'n =
    'Blood Oxygen Level (%)'n
    'Heart Rate (BPM)'n
    'Sleep Duration (hours)'n
    'Step Count'n
    'Stress Level'n
  / link=glogit;
  score data=prelib.test
    out=prelib.model_score;
run;

```

Training the model using train dataset, then evaluating models score using test dataset

```

proc freq data=prelib.model_score;
  tables 'Activity Level'n * 'I_Activity Level'n;
run;

```

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of Activity Level by I_Activity Level			
	I_Activity Level(Into: Activity Level)			
	Activity Level	Active	Highly_Active	Sedentary
	Active	405 14.11 42.36 33.53	133 4.63 13.91 32.52	418 14.56 43.72 33.33
	Highly_Active	390 13.58 42.62 32.28	129 4.49 14.10 31.54	396 13.79 43.28 31.58
	Sedentary	413 14.39 41.30 34.19	147 5.12 14.70 35.94	440 15.33 44.00 35.09
	Total	1208 42.08	409 14.25	1254 43.68
	Frequency Missing = 49			

```

data prelib.model_score;
    set prelib.model_score;

    correct = ('Activity Level'n = 'I_Activity Level'n);
run;

proc means data=prelib.model_score mean;
    var correct;
run;

```

Models accuracy

The MEANS Procedure

Analysis Variable : correct	
	Mean
	0.3335616

Conclusion:

The model shows poor ability to distinguish between Activity Level categories, as predictions are widely distributed across all classes with significant overlap. This indicates weak separability between the classes based on the available features.

To handle missing values in the Activity Level variable, mode imputation was applied, where the most frequent category (Sedentary) was used to replace missing observations.

```

proc freq data=prelib.imputed_missing_values_datanona;
    tables 'Activity Level'n;
run;

```

The FREQ Procedure

Activity Level	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Active	3186	33.40	3186	33.40
Highly_Active	3122	32.73	6308	66.12
Sedentary	3232	33.88	9540	100.00
Frequency Missing = 195				

```

data prelib.smartwatch_dataset_cleaned;
  set prelib.imputed_missing_values_datanona;

  if missing('Activity Level'n)
  then 'Activiy Level'n = 'Sedentary';

run;

```

The FREQ Procedure

Activity Level	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Active	3186	32.73	3186	32.73
Highly_Active	3122	32.07	6308	64.80
Sedentary	3427	35.20	9735	100.00

The MEANS Procedure

Variable	N Miss
User ID	0
Heart Rate (BPM)	0
Blood Oxygen Level (%)	0
Step Count	0
Stress Level	0
Sleep Duration (hours)	0

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